



Semester II Examinations 2018/2019

Exam Code(s)	3BME1, 3BMS2, 3BS9, 4BMS2, 4BS2, 4FM2
Exam	3rd Arts, 3rd Science, 4th Arts, 4th Science
Module	Numerical Analysis II
Module Code	MA378
External Examiner(s)	Professor T. Brady
Internal Examiner(s)	Professor G. Ellis Dr N. Snigireva*
Instructions	Answer THREE questions. All questions carry equal marks.
Duration	2 hours
No. of Pages	4 pages (including this cover page)
Discipline	Mathematics
Requirements:	
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical Tables/ Log Tables	Yes

1. Let p_n be the polynomial of degree n that interpolates a function f at the $n + 1$ distinct points $\{x_0, x_1, \dots, x_n\}$.
 - (a) Show how to construct such a polynomial. [4 marks]
 - (b) Show that there is only one such p_n . [4 marks]
 - (c) State and prove Cauchy's theorem for $f(x) - p_n(x)$. [12 marks]

- (d) Let p_2 be the polynomial that interpolates $f(x) = \cos(x/4)$ at the points $\{1, 2, 3\}$. Use Cauchy's theorem to give an upper bound for the error, $|f(x) - p_2(x)|$, at $x = 2.4$. [10 marks]

2. Let l be the piecewise linear spline interpolant to f on $n + 1$ equally spaced points $a = x_0 < x_1 < \dots < x_n = b$. Let $h = (b - a)/n$.

- (a) Use Cauchy's theorem to give an upper bound for $\max_{a \leq x \leq b} |f(x) - l(x)|$ in terms of h and f'' . [9 marks]

- (b) Now consider $f(x) = \frac{1}{1 + x^2}$ on $[-5, 5]$.

- (i) For $n = 2$ write down the piecewise linear spline l that interpolates f . [6 marks]

- (ii) Is it true that

$$\lim_{n \rightarrow \infty} \max_{-5 \leq x \leq 5} |f(x) - l(x)| = 0 \quad ?$$

Justify your answer. [4 marks]

- (c) (i) Fix a set of interpolation points and let L be the operator that maps a continuous function f to its linear spline interpolant l . Explain why $L(L(f)) = L(f)$. [2 marks]
- (ii) Let $\hat{l}$ be any piecewise linear approximation to a given function f on a fixed set of interpolation points, and l be the piecewise linear interpolant to f . Show that

$$\max_{x_0 \leq x \leq x_n} |f(x) - l(x)| \leq 2 \max_{x_0 \leq x \leq x_n} |f(x) - \hat{l}(x)|.$$

Explain what is the importance of this result. [9 marks]

3. (a) (i) Explain what is meant by Newton-Cotes quadrature.

[4 marks]

- (ii) Derive the 3-point Newton-Cotes quadrature rule, that is, Simpson's rule.

[8 marks]

- (iii) Use Simpson's rule to estimate $\int_0^5 e^x \cos(\pi x/2) dx$.

[4 marks]

- (iv) Is it true that $\lim_{n \rightarrow \infty} Q_n(f) = \int_a^b f(x) dx$?

Justify your answer.

[2 marks]

- (b) Let $\{\tilde{p}_i\}_{i=0}^\infty$ be the sequence of orthogonal monic polynomials with respect to the inner product

$$(f, g) := \int_a^b f(x)g(x) dx$$

- (i) Prove that $\tilde{p}_i$ is orthogonal to all polynomials of degree less than i .

[4 marks]

- (ii) Prove that the zeros of $\tilde{p}_i$ are simple (that is, distinct).

[4 marks]

- (iii) Prove that the zeros of $\tilde{p}_i$ all lie in the interval $[a, b]$.

[4 marks]

4. (a) (i) Let $x_0, \dots, x_n$ be the distinct zeros of $\tilde{p}_{n+1}$, the $(n+1)$ th polynomial in the sequence of orthogonal monic polynomials $\{\tilde{p}_i\}_{i=0}^{\infty}$ with respect to the inner product

$$(f, g) := \int_a^b f(x)g(x) dx.$$

Set

$$G_n(f) = w_0 f(x_0) + \dots + w_n f(x_n)$$

where $\{w_i\}_{i=0}^n$ are quadrature weights. Show that $G_n(f)$ has precision $2n+1$. **[12 marks]**

- (ii) Show that it is impossible to choose $n+1$ quadrature points, x_k , and weights, w_k , so that the $n+1$ -point quadrature rule

$$\int_a^b f(x) dx \approx \sum_{k=0}^n w_k f(x_k)$$

has precision $2n+2$.

[6 marks]

- (b) Consider the following differential equation:

$$-u''(x) = x \text{ on } (0, 2), \text{ and } u(0) = u(2) = 0. \quad (1)$$

- (i) State the variational formulation of the differential equation in (1) and show that the solution to the variational problem is unique. **[8 marks]**
- (ii) Suppose one wanted to compute an approximation to (1) using the finite element method (FEM) with the usual piecewise linear Galerkin basis functions on the uniform mesh $\{x_0, x_1, \dots, x_n\}$. Give a formula for the basis functions. How is the FEM solution expressed in terms of this basis?

[4 marks]